

INDR 100

INTRODUCTION TO INDUSTRIAL ENGINEERING

Lecture #3

A Gallery Walk of Industrial Engineering Problems

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The Scientist vs. The Engineer

- **The Scientist asks:** 'Why does the apple fall?' (Knowledge)
- **The Engineer asks:** 'How can I use gravity to power a city?' (Application)



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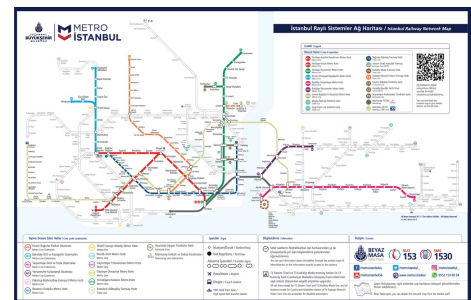
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We don't just study the world. We optimize it.

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The Map is Not the Territory

- Reality is messy. Models are clean.
- A model ignores details (color of train, name of driver).
- A model focuses on variables (routes, stops, time).
- Several models that changed the world.

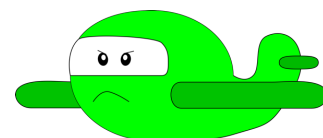
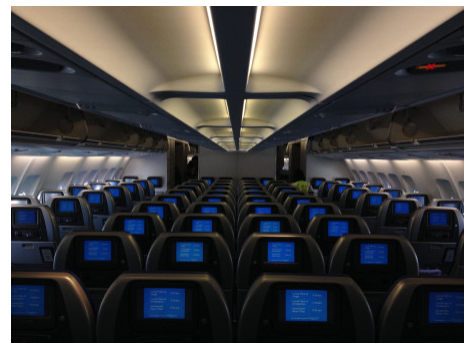


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How much to buy? (Newsvendor)

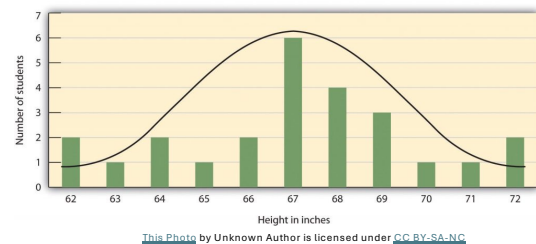
- Scenario: Buying plane tickets to resell.
- Risk 1: Buy too many = WASTE (Empty seats).
- Risk 2: Buy too few = SHORTAGE (Lost customers).
- Question: What is the 'Magic Number'?



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The Crystal Ball

- We use the past to predict the future.
- We don't just guess.
- We calculate the Probability of demand.



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How much to buy?

- A travel booking company buys blocks of plane tickets from an airline **before knowing exact demand**. They plan to resell them to customers.
- **Uncertainty:** The company doesn't know how many travelers will want tickets for that route.
- **Risks:**
 - If they **buy too few tickets**, they sell out quickly → missed sales and disappointed customers.
 - If they **buy too many tickets**, not all are sold at full price → they must offer last-minute discounts or promotions to recover part of the cost.
- **Decision:** How many tickets should they buy?

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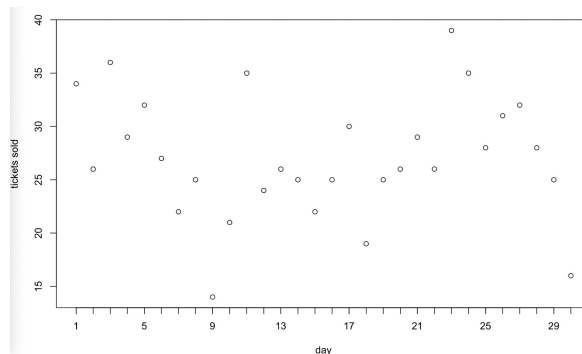
Buy or not to buy: that is the question

- **Purchasing cost per ticket: 2,000 TL:**
- **Selling cost per ticket: 2,500 TL**
- **Salvage value per ticket: 1,800 TL**
- **Data (sales for the last 30 days):**
 19 21 23 18 18 24
 27 29 25 29 19 16
 32 25 18 23 27 18
 20 14 24 26 31 31
 16 25 32 24 18 20

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Buy or not to buy: that is the question

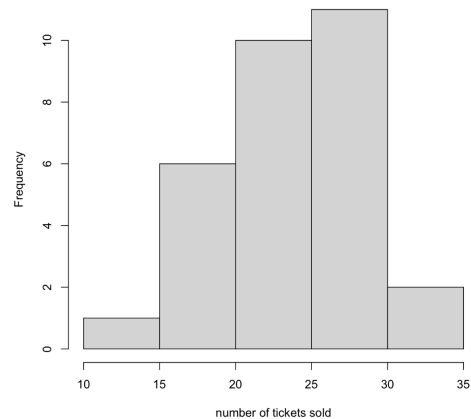
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Buy or not to buy: that is the question

- Purchasing cost per ticket: 2,000 TL:
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Buy or not to buy: that is the question

$$\begin{aligned} \max \quad & x \\ \text{s.t.} \quad & F(x)(s - c) + (1 - F(x))(p - c) \geq 0 \end{aligned}$$

Solution:

$$\Rightarrow F(x)[(s - c) - (p - c)] + (p - c) \geq 0$$

$$\Rightarrow F(x)(s - p) + (p - c) \geq 0$$

$$\Rightarrow (p - c) \geq F(x)(p - s)$$

$$\Rightarrow F(x) = \frac{p - c}{p - s} \quad (\text{at optimum})$$

$$\Rightarrow x^* = F^{-1}\left(\frac{p - c}{p - s}\right).$$

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What if?

- There is a **trend in the demand**
- The unsold items can be stored for the next round with an **inventory cost**
- We want to limit the **risk of losing money**?
- Demand depends on the price you set?
- The decision maker orders multiple products that share a **budget or capacity constraint**
- The amount received from a supplier is random (e.g., agriculture, manufacturing yield losses).
- Multiple firms face the same market and interact through **price or quantity competition** (Cournot/Bertrand settings).
- A central warehouse allocates inventory across multiple stores facing random demand.